

Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: Diagnosing Class-Conditional Negative Transfer

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Abstract

Sarcasm detection requires intent beyond literal wording. Large language models (LLMs) can expand sarcasm resources, but evaluation must distinguish sarcasm-relevant pragmatic cues from source-related ones. A shared encoder supports both sarcasm detection and machine-generated text detection (MGTD), yet source-predictive cues may distort sarcasm predictions. We harmonize Chinese and English resources and construct DualTest for source $\times$ sarcasm subgroup analysis, with prompt-specified style serving as the sarcasm label for AI-generated texts. On AI-generated texts prompted to be non-sarcastic, the final plain joint model’s false-positive rate exceeds that of its sarcasm-only counterpart by 67.0 percentage points, a pattern we call *class-conditional negative transfer*. Scores shift upward despite substantial within-AI discrimination. Both languages show the pattern, which persists when unchanged detectors evaluate a second test set generated by another LLM. Representation diagnostics link source-related scores and directions to the affected sarcasm decisions and show that removing one data-derived high-variance direction changes the subgroup gap. Exploratory analysis finds that predicted human-authorship probabilities do not consistently track LLM-judge ratings of human-likeness obtained without ground-truth source labels. These findings motivate source $\times$ sarcasm subgroup evaluation and external validation before detector probabilities are reused as quality scores or rewards.

1 Introduction

Sarcasm expresses an intended attitude that may differ from an utterance’s literal meaning, so automatic detection depends on implicit contrast, communicative intent, and context. Errors can reverse sentiment predictions and disrupt opinion-mining or content-moderation systems (Tay et al., 2018; Vidgen et al., 2019). Datasets also differ

in how sarcastic texts are elicited or collected and whose judgments define the labels; these choices, together with domain, style, and language, affect annotation and model behavior (Oprea and Magdy, 2020; Casola et al., 2024; Jang and Frassinelli, 2024).

Large language models (LLMs) offer a way to expand sarcasm resources, but their usefulness depends on realizing the requested style without turning source-specific regularities into shortcuts. Evaluation should therefore separate source and sarcasm: both human-authored and AI-generated texts may be sarcastic or non-sarcastic. We refer to the four resulting combinations—human-authored sarcastic, human-authored non-sarcastic, AI-generated sarcastic, and AI-generated non-sarcastic texts—as source $\times$ sarcasm subgroups.

A shared encoder with task-specific heads can reuse one representation for both predictions, but it also creates a route for interference: machine-generated text detection (MGTD) favors features that distinguish human-authored from AI-generated text (Guo et al., 2023; Wang et al., 2024), whereas sarcasm detection should reflect intent rather than source. The labels themselves do not conflict; the risk lies in how the tasks organize the shared representation. If source-predictive cues influence the sarcasm head, information useful for MGTD but spurious for sarcasm can produce task conflict and negative transfer (Mueller et al., 2022; Hu et al., 2022b). This motivates testing how joint training changes sarcasm predictions.

Because source-predictive regularities and sarcasm cues vary across languages and corpora (Jang and Frassinelli, 2024; Casola et al., 2024; Wang et al., 2024), we examine the transfer pattern separately in Chinese and English.

In this paper, we construct DualTest, a bilingual evaluation set for studying joint MGTD–sarcasm training with separate source and sarcasm labels across these four subgroups. For AI-generated

texts, the sarcasm label records prompt-specified style. Using DualTest and same-configuration sarcasm-only controls, we show that joint training yields sharply higher false-positive rates on AI-generated texts prompted to be non-sarcastic. We call this pattern *class-conditional negative transfer* because the degradation is concentrated in one source $\times$ sarcasm subgroup rather than distributed uniformly across DualTest. The effect appears in both languages and persists when the same detectors evaluate outputs from another generator without retraining.

To connect decodable information with model behavior beyond probe accuracy, we use linear probes to test whether source can be recovered from a representation, compare MGTD- and sarcasm-head scores to test whether their decisions move together, and intervene on frozen representations by removing selected directions before reapplying the original sarcasm head (Hewitt and Liang, 2019; Elazar et al., 2021). Source-related scores and directions co-vary with the affected sarcasm decisions, and removing one data-derived high-variance direction changes the subgroup gap.

As a supplementary exploratory analysis, we examine whether detector probabilities agree with separately obtained ratings of sarcasm and human-likeness. This matters because classifier scores may be reused as evaluation or reward signals for generated text. We therefore compare detector sarcasm and human-authorship probabilities with LLM-judge ratings obtained without ground-truth source labels.

Contributions. We construct DualTest from Chinese and English resources and use same-configuration controls to identify class-conditional negative transfer concentrated on AI-generated texts prompted to be non-sarcastic. The effect persists across both languages and when the same detectors evaluate another generator’s outputs without retraining, while representation diagnostics link source-related scores and directions to the affected decisions and reveal sensitivity to a residual high-variance direction estimated from the fitting split, motivating semantic characterization and targeted causal tests. An exploratory comparison with LLM-judge ratings further motivates external validation before detector probabilities are interpreted as measures of sarcasm quality or human-likeness.

2 Related Work

2.1 Sarcasm Detection, Synthetic Data, and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. Performance nevertheless varies substantially across datasets, which differ in label definitions, domains, languages, and modalities. Label schemes may reflect intended or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024), while multimodal resources may use modality-specific annotations (Yue et al., 2024).

Recent studies evaluate LLM-generated sarcasm through predictive performance, diversity, linguistic properties, and human judgments (Majer et al., 2026), and associate synthetic-data characteristics with shortcut learning and poor transfer to human-authored text (Mahato et al., 2026). These findings motivate testing whether MGTD-derived source cues interfere with sarcasm detection in a shared encoder.

2.2 Machine-Generated Text Detection and Authorship Verification

MGTD includes likelihood-based zero-shot methods (Mitchell et al., 2023) and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). M4 documents generalization gaps across generators, domains, and languages (Wang et al., 2024), whereas RAID stress-tests unseen generators, decoding variations, and attacks (Dugan et al., 2024). These findings are consistent with reliance on generator- or domain-specific source cues, which may also shape a shared representation used for sarcasm detection. Bevendorff et al. (2025) frame realistic LLM detection as open-set authorship verification rather than closed-set attribution; here, binary human-versus-AI classification serves as a controlled auxiliary task.

To the best of our knowledge, the interaction between sarcasm detection and MGTD under shared-encoder joint training has not been directly examined. Because source is the MGTD target but should not determine sarcasm, we test whether the shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, transfer depends on task relatedness and data-distribution alignment (Zhang et al., 2023; Wu et al., 2020), while gradient conflict can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Matched single- and multi-task comparisons show similar failures and suggest that models may infer task identity from input distributions (Mueller et al., 2022); other work analyzes functional specialization in multi-task Transformers (He and Choi, 2021; Li et al., 2023). A complementary shortcut-learning account is that models exploit training-predictive features that fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). Shared representations can strengthen such correlations (Hu et al., 2022b; Chai et al., 2025), while additional datasets can introduce them and reduce worst-group performance (Compton et al., 2023).

Mitigation approaches either isolate task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modify information flow within a shared graph (Wu et al., 2019; Pilault et al., 2021). Our comparison of a plain shared encoder with two intermediate-representation variants follows the second direction and examines subgroup variation in negative transfer.

2.4 Detector Outputs as Evaluation and Reward Signals

Detector probabilities are sometimes reused as evaluation scores or generation rewards. For example, ViSP (Wang et al., 2026) includes a DIP-derived sarcasm score (Wen et al., 2023) in its PPO reward (Schulman et al., 2017). Such reuse assumes that a classifier’s continuous output tracks the intended quality dimension, an assumption not established by classification accuracy; optimizing a model-based proxy can also diverge from an external measure (Gao et al., 2023). As LLM judges are increasingly used to evaluate generated text (Zheng et al., 2023), we examine whether detector sarcasm probabilities agree with separately obtained LLM-judge sarcasm ratings. We also compare predicted human-authorship probabilities with judge-rated human-likeness because source classification and perceived human-likeness are related but distinct constructs.

3 Dataset Construction

Testing cross-task interference requires separate sarcasm and source labels, where source denotes human versus AI authorship rather than corpus. We pool the English iSarcasmEval data (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). Since none provides both labels, we map their annotations to a common binary sarcasm space, yielding 44,862 instances (approximately 87% English). From these corpora, we construct SarcTrain for sarcasm supervision, AuthTrain for source supervision, DualTest for source $\times$ sarcasm evaluation, and GenTrain for generation fine-tuning (Appendix A).

SarcTrain. We reserve 2,103 eligible sarcastic records for GenTrain, then sample SarcTrain from the remaining 42,759 records. It contains 10,000 human-authored texts for sarcasm classification, balanced across four language $\times$ sarcasm strata with 2,500 texts per stratum. Three strata are downsampled; the Chinese sarcastic stratum is expanded from 2,396 to 2,500 by sampling 104 records with replacement. We assign five folds after oversampling, so some duplicates cross folds (Appendix A).

AuthTrain. AuthTrain starts from 9,999 approximately language-balanced SarcTrain seeds (5,000 English; 4,999 Chinese) after seed-level quality control excludes one Chinese instance. Depending on the corpus, a seed is a topic, dialogue context, non-sarcastic rewrite input, or the text itself. The locally deployed Ollama model deepseek-r1:8b (Guo et al., 2025; Ollama, 2025) generates AI counterparts with four language $\times$ style templates, with style selected from each seed’s sarcasm label. Generation and quality control retain 9,996 counterparts (three unmatched seeds; 99.97% coverage), yielding 19,995 source-classification texts. Source and prompt-specified style are thus distinct, approximately balanced factors for source $\times$ sarcasm subgroup analysis (Appendix A).

DualTest. DualTest contains 4,664 texts for source $\times$ sarcasm evaluation. Its human subset comprises the official iSarcasmEval (1,400 English) and ToSarcasm (972 Chinese) test sets, which are excluded from the 44,862-record pool.

Generation and quality control retain counterparts for 2,292 of 2,372 human texts (96.6% coverage). Human texts retain dataset labels; sarcasm labels for AI-generated texts record prompt-specified style rather than independent human ratings of perceived sarcasm. The four subgroups contain 823 human-sarcastic and 1,549 human-non-sarcastic texts, plus 779 AI-generated texts prompted to be sarcastic and 1,513 prompted to be non-sarcastic. Record-level checks find no SarcTrain or AuthTrain overlap with either side of DualTest (Appendix A, which also gives the language $\times$ source $\times$ sarcasm distribution).

GenTrain. Separately, GenTrain contains 2,428 texts for the supervised fine-tuning (SFT) used in §6. From 5,257 eligible sarcastic records, we sample 2,103 (40%; 1,214 English and 889 Chinese) and add 325 Chinese records sampled with replacement, yielding 1,214 instances per language. It contains 374 rewriting, 839 response, and 1,215 comment instances (Appendix A).

4 Joint and Single-Task Models

4.1 Tasks and Architecture

Given a Chinese or English text x , the model predicts binary sarcasm and source labels, with sarcastic and human-authored as the positive classes; all four combinations are valid. The tasks share a DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how joint training changes the representation read by the sarcasm head. We compare three representation configurations: (1) plain sends [CLS] to both heads; (2) interaction-projection adds residual maps between task branches; and (3) RCNN adds recurrent and convolutional pooling. All retain the same encoder and task definitions (Appendix B).

First, the plain configuration serves as the shared-encoder baseline and sends [CLS] directly to both task heads.

Second, interaction-projection forms 256-dimensional sarcasm and MGTD vectors h_s and h_a , updated as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$, where $P_{a \rightarrow s}$ and $P_{s \rightarrow a}$ are learned affine maps in the arrow directions; subscripts name branches and primes mark updated vectors. Each branch supplies one key-value vector, so the attention weight equals one and each P composes the corresponding value and output projections.

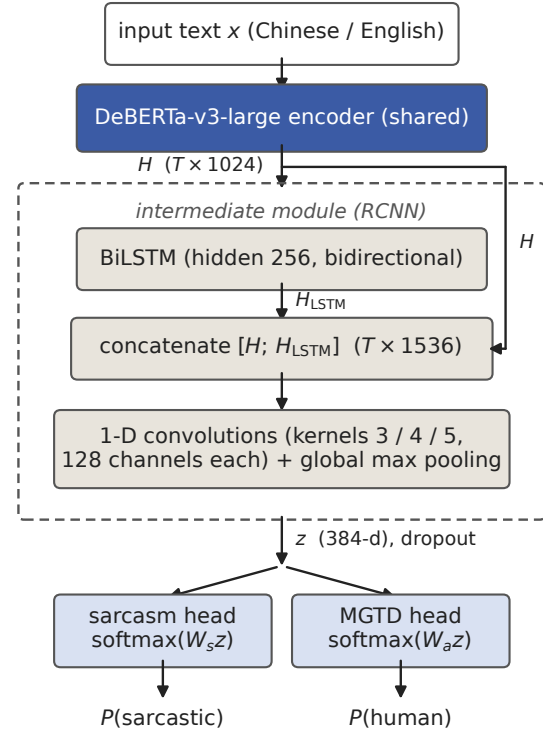


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with two learned residual projections between the task branches, while retaining the same encoder backbone and task definitions (Appendix B).

Finally, RCNN applies a one-layer BiLSTM, parallel one-dimensional convolutions, and global max pooling, a text-classification design (Lai et al., 2015) used for multilingual sarcasm detection (Yang and Ikeda, 2024). With 256 hidden units per direction, the BiLSTM yields a 512-dimensional vector per token, concatenated with the 1,024-dimensional encoder output. Convolutions of widths 3, 4, and 5 with 128 channels each and global max pooling produce a 384-dimensional vector z , followed by dropout and the two heads. Figure 1 illustrates this configuration.

4.2 Training Protocol

We distinguish the three representation configurations from the three training procedures: joint, sarcasm-only, and MGTD-only. Joint models have two heads; task-specific controls activate one. We compare joint and sarcasm-only procedures for plain and RCNN, holding the encoder, configura-

tion, sarcasm data, folds, and sarcasm hyperparameters fixed. Joint training additionally uses AuthTrain batches, shared updates from both tasks, a mixed schedule, and joint early stopping. Compute exposure differs: the control traverses the full sarcasm fold each epoch, whereas joint training samples sarcasm batches within the mixed schedule. The contrast therefore covers the complete joint-training procedure.

Training alternates task-homogeneous batches at a 2:1 MGTD:sarcasm ratio, approximating the dataset sizes. Let $z_s(x)$ and $z_a(x)$ be the head inputs and W_s and W_a their classifiers; s and a denote sarcasm and authorship. Plain and RCNN share one representation, whereas interaction-projection uses h'_s and h'_a . For active task t , B_t contains input-label pairs (x, y) and incurs only its task loss:

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z_s(x), y), & B_t \subseteq \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z_a(x), y), & B_t \subseteq \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is weighted sarcasm cross-entropy with $w = [1.0, 2.5]$ for the non-sarcastic and sarcastic classes, and CE is unweighted MGTD cross-entropy. We fix the weights before DualTest evaluation and share them across matched procedures. Both losses use logits; softmax is applied only for probability-based evaluation. Each batch updates the active head and shared parameters. The encoder uses a lower learning rate than new modules, the sarcasm-head rate is fixed across configurations, and Appendix B reports the MGTD-head rates. Binary predictions use a fixed 0.5 threshold unless stated otherwise.

For dual-task models, checkpoint selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics. Single-task controls use their own validation loss with the same five-epoch patience (Appendix B).

We use five fold-matched runs. In run k , four SarcTrain folds train the sarcasm task and the fifth provides validation; all runs share a fixed AuthTrain 90/10 split, whose approximately 2,000-example holdout provides a common MGTD validation set. Each run reloads its configuration’s pre-trained encoder and newly initializes the remaining modules and heads; all configurations use the same folds and seed schedule. DualTest is held out until evaluation. We report mean $\pm$ standard deviation across runs to summarize variation from the sarcasm fold and initialization; for subgroup rates from separately trained final models, Wilson intervals summarize uncertainty over DualTest examples (Table 3; Appendix C.3).

For each configuration used in the subgroup analysis, we train a final model on all applicable training data for a fixed epoch count determined from the five validation runs, using an independent seed and no early stopping. These models provide the subgroup and representation analyses but are excluded from the five-run aggregates; the exploratory analysis in §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

We organize the analysis in four stages: system comparison, matched joint-single-task comparison, source $\times$ sarcasm subgroup evaluation across languages and generators, and representation-level diagnosis of the resulting error pattern.

5.1 System-Level Comparisons

Backbone comparison. Under the same protocol, we compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023); for DeBERTa-v3-large, we also test interaction-projection and RCNN representation pathways. Table 1 reports positive-class F1 for both tasks.

On DualTest, the DeBERTa configurations have higher mean F1 than mBERT and XLM-R, with RCNN highest on both tasks (0.6309 sarcasm; 0.9301 MGTD). Subsequent analyses focus on the DeBERTa-v3-large configurations, treating the plain shared encoder as the reference and interaction-projection and RCNN as alternative representation pathways. Complete results are in Appendix C.

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLm-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.003
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction-projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and representation configurations (DualTest; five-run mean $\pm$ standard deviation; bold indicates the highest mean).

Config.	Task	Joint	Single	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	-6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	-4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Joint vs. corresponding single-task procedures (mean $\pm$ standard deviation over five fold-based runs; Δ = joint – single-task, in percentage points).

Joint versus single-task training. We compare the plain and RCNN joint procedures with matched single-task controls. Each pair holds architecture and task-specific data fixed; joint training additionally updates the shared encoder with the other task.

Sarcasm F1 is lower under joint training in all five matched runs, by 6.08 pp for plain and 4.43 pp for RCNN on average; MGTD F1 is higher by 1.67 and 0.49 pp. Joint training therefore produces a consistent aggregate sarcasm loss, smaller under RCNN. We next locate this loss across source $\times$ sarcasm subgroups.

5.2 Class-Conditional Negative Transfer

To locate the aggregate loss, we evaluate the separately trained final models across the four source $\times$ sarcasm subgroups. Sarcastic subgroups use recall and non-sarcastic subgroups use false-positive rate (FPR); human texts retain dataset annotations, while sarcasm labels for AI-generated texts record prompt-specified style. Plain and RCNN provide matched joint–sarcasm-only comparisons, and interaction-projection provides an additional joint representation pathway (Table 3); MGTD-head results are in Appendix C.

Loss concentration. On AI-generated texts prompted to be non-sarcastic, FPR rises from 19.3% under sarcasm-only training to 86.3%

under plain joint training (67.0 pp); the matched RCNN increase is 14.8 pp (19.3% to 34.1%). The plain-model increase on human-authored non-sarcastic texts is only 4.3 pp. The loss is therefore concentrated in the AI-generated non-sarcastic subgroup, a pattern we call class-conditional negative transfer.

Discrimination and score shift. Within AI-generated texts, the area under the receiver operating characteristic curve (AUC) for sarcasm ranges from 0.82 to 0.88 across models. Treating prompted-sarcastic texts as positives, AUC can be interpreted as the probability that a randomly selected prompted-sarcastic text receives a higher predicted-sarcasm score than a randomly selected prompted-non-sarcastic text, with ties counted as one half. Thus, the models retain substantial threshold-free ranking ability. In the AI-generated non-sarcastic subgroup, the plain model’s median predicted sarcasm probability rises from 0.204 under sarcasm-only training to 0.767 under joint training; the joint model’s interquartile range, 0.635–0.839, lies above the 0.5 decision threshold. The high FPR is therefore consistent with an upward score shift rather than collapsed within-AI ranking (Appendix C.3).

5.3 Cross-Language and Cross-Generator Recurrence

The subgroup effect has the same direction in both languages. Joint–sarcasm-only FPR increases for English and Chinese are 63.5 and 79.4 pp for plain, and 16.9 and 7.7 pp for RCNN (Appendix C.3; Table 10).

For generator transfer, unchanged detectors evaluate Qwen3.5-9B outputs produced with the same language $\times$ style templates (Ollama tag qwen3.5:9b) (Qwen Team, 2026). On non-sarcastic outputs, plain joint and sarcasm-only FPRs are 91.3% and 7.9% (83.4 pp, versus 67.0 pp on DeepSeek), while matched RCNN rates are 22.1% and 10.6%. The higher joint-training FPR therefore recurs on a second generator without detector retraining (Figure 2; Appendix C.3; Table 9).

5.4 Representation-Level Diagnosis

For each DualTest text, the frozen final plain joint model yields a single 1,024-dimensional pooled vector h summarizing the whole text. Q1 also uses vectors from the matched sarcasm-only model; Q2

Model	H sarc. recall	H non-sarc. FPR	AI sarc. recall	AI non-sarc. FPR
Plain dual-head	67.1%	32.9%	98.2%	86.3%
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9%
RCNN dual-head	80.8%	40.5%	89.3%	34.1%
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3%
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3%

Table 3: Sarcasm predictions across the four source $\times$ sarcasm subgroups. Human texts use dataset annotations; sarcasm labels for AI-generated texts record prompt-specified style. Dual-head rows use joint training, and single-head rows are sarcasm-only controls. Bold marks the focal AI-generated non-sarcastic subgroup; confidence intervals and paired tests are in [Appendix C.3](#).

and Q3 use the joint-model h . An intervention changes h before the unchanged sarcasm head is reapplied; the encoder and head remain frozen. The questions address source availability, association with sarcasm scores, and false-positive allocation. [Appendix E](#) gives details.

Q1: Is source available? A linear probe tests whether an attribute can be recovered from frozen representations without changing the model ([Hewitt and Liang, 2019](#)). For each model, we fit logistic-regression probes to all 4,664 DualTest vectors and report stratified five-fold ROC AUC over the human-versus-AI labels (0.5 is chance; 1.0 is perfect ranking). AUC is 0.986 for joint-model vectors and 0.974 for sarcasm-only vectors, so source is highly linearly readable in both. The latter result shows that this decodability is not unique to MGTD supervision: systematic human–AI differences are represented even when source is not a training label.

Q2: Is source associated with sarcasm decisions? A logit is a pre-softmax class score. On AI-generated non-sarcastic texts, MGTD AI and sarcasm logits correlate positively ($\rho = 0.519$): higher AI logits tend to accompany higher sarcasm logits. Correlations within length terciles (0.44, 0.56, 0.63) show that coarse length differences alone do not explain this. A *direction* is a vector in h -space; for a linear head, the class-weight difference indicates how h can move to raise one class logit relative to the other. Thus, $d_{\text{sarc}} = W_s[1] - W_s[0]$ raises the sarcastic-over-non-sarcastic logit, and $d_{\text{AI}} = W_a[0] - W_a[1]$ raises the AI-over-human logit (MGTD class 1 is human). The source-probe direction is the normalized coefficient of a separate fitting-half logistic probe, distinct from the full-set probes in Q1.

Ordinary cosine weights dimensions equally and finds the head directions nearly orthogonal ($\cos \approx -0.03$). Covariance-aware alignment

is instead a data-weighted similarity between directions, emphasizing dimensions where observed representations vary more ([Ledoit and Wolf, 2004](#)). Its absolute similarity with the sarcasm direction is 0.49 for the MGTD-head direction and 0.41 for the source-probe direction. Thus, source-related scores co-vary with sarcasm scores, and the data-weighted similarities are markedly larger than ordinary cosine.

Q3: Do interventions change source-group false-positive allocation? DualTest is split 50/50 within each source $\times$ sarcasm subgroup. Label-dependent directions and transforms are estimated on the fitting half, fixed, and applied to the evaluation half. Direction removal centers h , subtracts its projection onto a direction or subspace, restores the mean, and feeds the modified vector to the unchanged sarcasm head ([Elazar et al., 2021](#)). These vectors form the transformed evaluation set; [Appendix E](#) gives operator-specific statistics.

The FPR gap—AI-generated minus human-authored FPR on non-sarcastic texts—shows how the unchanged sarcasm head allocates errors. For each transformed set, a newly cross-validated linear probe yields residual source AUC, measuring how much human-versus-AI information remains linearly readable. The untouched within-model baseline is a 52.0-pp gap with source AUC 0.982, distinct from the earlier 67.0-pp joint–sarcasm-only difference.

(a) Source-direction interventions. To test whether a single learned source contrast controls the gap, we remove either the joint model’s MGTD-head direction or the fitting-half source-probe direction. Their gaps are 56.0 and 51.9 pp, with source AUCs of 0.982 and 0.981. To target source information distributed across multiple linear directions, we apply INLP, which repeatedly fits a source classifier and removes its predictive

direction to construct a multidimensional source-targeted subspace (Ravfogel et al., 2020). Removing 16 INLP directions gives a 72.1-pp gap and source AUC 0.974. These removals leave source highly readable and do not narrow the gap.

(b) Covariance-aware intervention. We apply LEACE, a closed-form linear transform designed to reduce linearly decodable source information while accounting for representation covariance (Belrose et al., 2023). LEACE lowers source AUC from 0.982 to 0.589 and the gap from 52.0 to 19.1 pp, whereas ten covariance-matched random directions leave gaps of 51.4–52.7 pp. The two groups’ unweighted mean FPR remains near 59% (58.8 to 59.1) because human non-sarcastic FPR rises from 32.8 to 49.5. LEACE therefore changes how the frozen sarcasm head allocates false positives between sources.

(c) Study-specific residual-geometry intervention. To explore dominant geometry outside a low-dimensional source subspace, we use three INLP directions to define a study-specific residual space and extract its PC1 (*INLP-3 residual PC1*). On the fitting half, we (i) fit three orthonormal INLP source directions, (ii) project centered representations onto their orthogonal complement, (iii) run PCA on the residual representations, and (iv) retain the first, highest-variance component. On the evaluation half, we remove only this PC1 from each original representation; the three INLP directions define the search space but are not removed. This intervention narrows the gap to 12.6 pp while source AUC remains 0.982. The FPR gap therefore does not consistently track residual linear source AUC, so source decodability alone is insufficient to explain the frozen head’s source-group error gap.

For Q3, LEACE and INLP-3 residual PC1 narrow the source-group FPR gap, unlike the tested head, probe, and INLP removals, which neither narrow it nor substantially reduce residual source AUC. Together, Q1–Q3 show source availability, association with sarcasm scores and data-weighted directions, and sensitivity of false-positive allocation to representation geometry (Appendix E.4; Table 19).

6 Exploratory Comparison of Detector Scores with LLM-Judge Ratings

We compare RCNN’s human-authorship and sarcasm probabilities with human-likeness and

sarcasm ratings because detector scores may be reused for quality evaluation or rewards. DeepSeek-V4-Flash (DeepSeek, 2026), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025) rate paired outputs from base Qwen3-8B (Yang et al., 2025) and its GenTrain-adapted QLoRA counterpart (Dettmers et al., 2023) for 300 prompts (180 Chinese topic comments and 120 English rewrites). Ratings omit source labels and cover 600 outputs; we compare condition means and text rankings.

Mixed agreement by setting and level. In Chinese generation, detector sarcasm probability rises from 0.780 to 0.895 while the three-judge mean falls from 3.99 to 3.80; both decline in English rewriting. Sarcasm ratings fall in all six setting–judge comparisons, five significantly after Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995). Predicted human-authorship rises in both settings while human-likeness ratings fall. Text-level sarcasm correlations are near zero in Chinese (−0.020) and moderately positive in English (+0.338); pooled human-likeness correlation is negative (−0.212), driven largely by opposing condition means. Thus, only English sarcasm agrees across both levels: Chinese sarcasm has opposite condition changes and near-zero rank agreement, while human-likeness has opposite changes and negative pooled rank agreement. Detector probabilities therefore require external validation before use as quality scores or rewards. Appendix C reports tests, correlations, and means (Table 12–14); Appendix D gives disagreement cases (Table 15–17).

7 Conclusion

We observe class-conditional negative transfer: on AI-generated non-sarcastic texts, the final plain joint model’s FPR exceeds its sarcasm-only counterpart by 67.0 pp, with upward scores despite substantial within-AI discrimination. The pattern recurs across languages and on Qwen3.5 outputs; RCNN and interaction-projection have lower subgroup rates. Interventions show that error allocation depends on representation geometry beyond the MGTD direction or residual linear source AUC, motivating semantic and causal analysis. Predicted human-authorship also diverges from human-likeness ratings, requiring external validation before these probabilities serve as quality scores or rewards.

Limitations

DualTest combines human sarcasm annotations with prompt-specified style labels for AI-generated texts. Accordingly, false positives for AI-generated texts are defined relative to the style specified in the prompt rather than independent human annotation of perceived sarcasm.

The core joint-versus-sarcasm-only comparisons evaluate the complete joint-training procedure. Within each pair, the architecture, sarcasm data, fold assignments, sarcasm-head learning rate, and class weights are held fixed, while MGTD batches, mixed-task scheduling, shared-parameter updates, and joint early stopping are introduced together. Compute- and update-matched controls could help disentangle these components. Comparisons across architectures do not isolate architecture effects because model capacity and optimization also differ. The cost-sensitive sarcasm loss may also affect absolute recall and false-positive rates.

Because oversampling precedes fold assignment, some duplicated SarcTrain records occur in different folds and may appear on both sides of a training-validation split, introducing dependence into checkpoint selection, although these records do not occur in DualTest. Generation failures from deepseek-r1:8b leave 80 human test texts without retained AI counterparts.

The representation analyses on one separately trained final model connect source-related structure to behavior but do not isolate a unique causal feature or characterize variation across seeds. Some diagnostics use unlabeled statistics from all DualTest vectors: covariance-aware alignment uses the full-set covariance, and specified Euclidean interventions use the full-set mean. They are therefore transductive diagnostics, not fully held-out intervention estimates. The intervention split also does not preserve human-AI seed pairs, and 24.7% of retained pairs cross the two halves, so the partitions are not fully independent at the seed level. The detector-judge comparison lacks human ratings, and language is not varied independently of generation task. Future work should use independent style annotations, matched training budgets, human ratings, and more generators.

Ethical considerations

All human-authored texts were obtained from publicly released research datasets: iSarcasmEval, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. The iSarcasmEval and CSC repositories display MIT licenses and the News Headlines v2 Kaggle release states CC BY 4.0, whereas we do not redistribute human-authored text from sources without an identified redistribution license ([Appendix A](#)). We collected no new user data or recruited participants, and we did not perform an additional audit of the released corpora for personally identifiable information or offensive content.

All AI-generated texts were produced locally using publicly released open-weight models (deepseek-r1:8b, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They were used only for training and evaluation and were labeled as AI-generated. For automated evaluation, evaluation inputs, generated outputs, and rating instructions were transmitted to the DeepSeek, Google, and OpenAI APIs. Dataset metadata and ground-truth source labels were not transmitted.

Given the observed subgroup error and the absence of validation for individual-text authorship attribution or high-stakes deployment, detector outputs should not be used as evidence of authorship or misconduct without human review and domain-specific validation. They should not be deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The error concentrated on AI-generated texts prompted to be non-sarcastic is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools supported language editing, structural revision, and formatting, but did not generate the reported experimental outputs or statistics. The authors determined the study design, analyses, and conclusions and take full responsibility for the work.

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A Detailed Dataset Construction, Generation, and Quality Checks

Corpus	Lang.	Inst.	Resource use
iSarcasmEval (SemEval-2022 T6)	English	3,467	SarcTrain; GenTrain; DualTest*
CSC	English	6,878	SarcTrain; GenTrain
ToSarcasm	Chinese	3,898	SarcTrain; GenTrain; DualTest*
News Headlines v2	English	28,619	SarcTrain
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	SarcTrain

Table 4: Input corpora and resource use. The instance counts refer to the 44,862-record pooled collection; *DualTest uses separate official test splits excluded from these counts. Genres and annotations are iSarcasmEval (tweets, speaker self-annotation), CSC (conversational responses), ToSarcasm (topic comments, third-party annotation), News Headlines (distant supervision from The Onion / HuffPost), and the Open Chinese Corpus (Weibo, third-party annotation).

A.1 Source corpora, sampling, and release scope. Table 4 lists the input corpora and how each is used. The pooled collection contains 44,862 instances in total (86.9% English and 13.1% Chinese). SarcTrain is balanced across the four language $\times$ sarcasm strata by down-sampling the two English strata and the Chinese non-sarcastic stratum, then sampling 104 additional Chinese sarcastic records with replacement. The Chinese sarcastic source pool contains 2,396 records corresponding to 2,394 distinct stored text values; after expansion to 2,500 records, its text-duplication rate is 4.24% (A.3).

The iSarcasmEval and CSC repositories display MIT licenses and the News Headlines v2 Kaggle release states CC BY 4.0, whereas we did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from the latter two sources. Accordingly, we treat only eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources as redistributable project artifacts.

A.2 Language $\times$ source $\times$ sarcasm distribution of DualTest. The retained counts are reported in Table 5.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human-authored texts use dataset sarcasm annotations; AI-generated texts use prompt-specified style labels.

A.3 Data quality checks. Automated checks confirm language and sarcasm-label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching finds that neither SarcTrain nor AuthTrain shares records with either the human or AI side of DualTest.

SarcTrain’s text-duplication rate is 0.34% for English and 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic stratum (4.24%; non-sarcastic stratum: 0.08%) and results primarily from the oversampling used to fill that stratum (A.1). Because oversampling precedes fold assignment, for 86 of the 104 added records, another copy originating from the same source record is assigned to a different fold. The two counting rules are summarized in Table 6.

Counting rule	Dup. sets	Records	Cross-fold
Exact stored text; over-sampled copies included	116	241	98
Case/whitespace normalized; added copies excluded	19	46	16

Table 6: Duplicate-record counts under the two counting rules used in the data-quality audit.

The first rule underlies the reported duplication rates; the second isolates pre-existing duplicates after the added copies are removed. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest.

A.4 AI generation protocol. The AI-generated texts in AuthTrain, the primary DeepSeek-generated DualTest AI subset, and the separate Qwen3.5 transfer subset are produced by locally deployed models through the Ollama API. The model deployed under the Ollama tag `deepseek-r1:8b` generates AuthTrain and the primary DualTest AI subset. Qwen3.5-9B, deployed under the tag `qwen3.5:9b`, generates the transfer subset evaluated with the same detectors without retraining. Both models use temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a maximum output length of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates constrain output style and length to 20–80 Chinese characters or 10–50 English words. These constraints are intended to reduce, but do not eliminate, source differences in length, so Appendix E separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its Hanyu Pinyin transliteration is:

[Renwu] Shengcheng yi tiao fengci
 fengge de pinglun
 [Zhuti] {seed}
 [Yaoqiu]
 - Zhijie geichu jianduan de pinglun;

buyao jieshi silu huo fenxi
 - Fengge yao fengci, youmo, kuazhang
 - Biaoda yu zhuti xiangguan de
 guandian ji ke; bubi zhuiqiu yanjin
 - 20--80 zi wei jia
 [Pinglun]

Its English translation is:

[Task] Generate a sarcastic comment
 [Topic] {seed}
 [Requirements]
 - Directly provide a short comment;
 do not explain your reasoning or
 analysis
 - The comment should be sarcastic,
 humorous, and exaggerated
 - It is sufficient to express a
 viewpoint related to the topic; the
 comment need not be rigorous
 - Preferably 20--80 Chinese
 characters
 [Comment]

The Chinese non-sarcastic template differs only in the style line, which requests a normal, natural style. The English templates use the same field structure and request either a sarcastic, humorous, exaggerated style or a normal, natural style, with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ní* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that still exceed the length limit or exhibit cyclic repetition after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-generated AI test subset has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use the full set of retained outputs for each generator rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation evaluation. After eligibility filtering, 5,257 records labeled as sarcastic remain for generation-task construction (2,972 English; 2,285 Chinese). After shuffling with seed 42, we sample 40% of these records, yielding 2,103 instances (1,214 English; 889 Chinese); language balancing then samples 325 addi-

tional Chinese records with replacement, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting instances, 839 response instances, and 1,215 comment instances. Generation evaluation uses a separately constructed set of 300 prompts that is not part of the GenTrain training-validation split (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. Language and generation task are therefore not independently varied. This set is used for detector scoring and all three LLM evaluations in §6.

B Hyperparameters and Training Details

Configuration	Enc.	Inter.	Sarc.	MGTD
Plain dual-head	5×10^{-7}	—	5×10^{-6}	1×10^{-5}
Plain	5×10^{-7}	—	5×10^{-6}	—
sarcasm-only				
Plain MGTD-only	5×10^{-7}	—	—	1×10^{-5}
Interaction-projection dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN	5×10^{-7}	5×10^{-6}	5×10^{-6}	—
sarcasm-only				
RCNN	5×10^{-7}	5×10^{-6}	—	1×10^{-5}
MGTD-only				

Table 7: Discriminative learning rates for the dual-head and task-specific controls. The plain setting is also used in the backbone comparison. A dash denotes an inactive module or task head.

B.1 Discriminative learning rates. Table 7 lists the per-module learning rates. The sarcasm-head learning rate is fixed at 5×10^{-6} in every configuration with an active sarcasm head. The MGTD-head rate is 1×10^{-5} for the plain dual-head model and the plain and RCNN MGTD-only controls, compared with 5×10^{-6} for the RCNN and interaction-projection dual-head models. Thus, the core joint-versus-sarcasm-only comparisons retain the same sarcasm-specific rates within each architecture. Because the RCNN joint-versus-MGTD-only comparison also changes the MGTD-head learning rate, it is reported descriptively rather than as an optimization-matched estimate of transfer; cross-configuration differences can likewise reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task’s own validation loss for the task-specific controls,

with a patience of five epochs in both procedures.

B.2 Shared training hyperparameters. Training uses `torch.optim.AdamW` with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. We implement a linear 200-step warmup followed by a constant learning rate using PyTorch’s `LambdaLR`. Training runs for at most 10 epochs, with an early-stopping patience of 5 and gradient clipping at 1.0. The physical batch size is 4 with gradient accumulation over 4 steps, giving an effective batch size of 16 for the active task. The MGTD:sarcasm sampling ratio is 2:1, and the sarcasm class weights are [1.0, 2.5] for the non-sarcastic and sarcastic classes, respectively. The task ratio and class weights were fixed before DualTest evaluation. The optimization settings reported here and in B.1 were selected based on prior practice and hardware constraints. We did not conduct a systematic hyperparameter search, and DualTest was not used to choose these optimization settings. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls process the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width- k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU. Global max pooling covers all convolution positions; no post-convolution padding mask is applied.

All fold-based runs use a seed equal to 42 plus the fold index and share the same fold assignments and random-seed values across architectures and ablations. The same fixed AuthTrain 90/10 split is reused in all five runs; only the SarcTrain validation fold changes. For each configuration, the epoch count used to train its final model is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. This mode-plus-one re-fitting rule was fixed before DualTest evaluation. The final model uses all applicable training data, a seed offset of +1000, no validation set, and no early stopping. Tables 1 and 2 report five-run means and standard deviations, whereas Table 3 and the subsequent fixed-model analyses use separately trained final models.

B.3 Model sizes and computational environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB; driver 570.124.04). Across the fold-based backbone, architecture, and ablation experiments and the training of final models, detector training and inference take approximately 170 GPU-hours on this single GPU. This figure excludes local text generation through Ollama, SFT (about two hours), and the representation analyses, which run on CPU. A single detector run takes from a few minutes to a few hours. The software environment is Python 3.11.0, PyTorch 2.9.0 (CUDA 12.8; cuDNN 9.10.2), transformers 4.57.1, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, pandas 2.3.3, peft 0.18.0, and datasets 4.4.1. AI-generated texts are produced locally through Ollama (Appendix A.4); LLM evaluation uses API calls made in May 2026 (Appendix C.4).

B.4 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Detrmers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA configuration (Hu et al., 2022a) uses rank 64, alpha 16, and dropout 0.0 and targets `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses a learning rate of 5×10^{-5} , three epochs, batch size 1, gradient accumulation 8, and a warmup ratio of 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint. Because language-balancing oversampling precedes the split, duplicated Chinese records may occur in both partitions.

C Complete Experimental Results

Sections C.1–C.3 supplement the detector and subgroup analyses in §5, while Sections C.4–C.8 provide protocols, controls, and additional results for the exploratory LLM-judge comparison in §6.

C.1 Macro-averaged F1 (supplement to Table 1). Under the same five-run protocol, the RCNN macro-averaged F1 scores computed from each run’s confusion matrix are 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sar-

casm macro-F1 is higher than its positive-class F1 (0.6309) because the non-sarcastic majority class has a higher F1. These values supplement the positive-class results in §5.1; because macro-F1 is not reported for every configuration, they are not used for cross-configuration ranking.

C.2 Complete source $\times$ sarcasm subgroup results for the MGTD head (including MGTD-only controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H $\times$ s	H $\times$ n	AI $\times$ s	AI $\times$ n
Plain dual-head	91.4	91.5	3.6	6.1
Interaction-projection dual-head	85.4	85.4	1.8	4.1
RCNN dual-head	94.8	93.2	5.5	6.6
Plain MGTD-only	90.4	86.6	4.1	4.2
RCNN MGTD-only	90.5	92.4	3.6	5.7

Table 8: MGTD-head results across the four source $\times$ sarcasm subgroups (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

In every subgroup, each MGTD-only control differs from its corresponding dual-head model by fewer than ten percentage points. No comparably large subgroup error appears on the MGTD side, consistent with §5.2.

When the same detectors evaluate the Qwen3.5-generated AI test subset without retraining, the MGTD head again shows no comparable subgroup increase: AI $\times$ non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for the plain, interaction-projection, and RCNN configurations; AI $\times$ sarcastic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.965.

C.3 Complete source $\times$ sarcasm subgroup results: confidence intervals, paired tests, language strata, and generator transfer (§5.2–§5.3). For each separately trained final model, we quantify uncertainty from the test instances with Wilson 95% intervals for subgroup rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. The instance-level bootstrap resamples text instances independently and does not preserve human–AI seed pairs. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the

between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3. Across the five final models, sarcasm AUC within AI-generated texts ranges from 0.82 to 0.88.

Figure 2 plots these subgroup rates for both generators, and Table 9 gives the corresponding intervals. For both matched model pairs, the false-positive rate remains higher under the joint procedure on the Qwen3.5-generated texts, although the magnitudes differ.

Model	DeepSeek	Qwen3.5
Plain dual-head	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Interaction-projection dual-head	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual-head	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single-head	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single-head	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 9: Wilson 95% intervals for the sarcasm false-positive rate on AI-generated texts prompted to be non-sarcastic (two generators; %).

Paired McNemar tests use predictions on the same DeepSeek-generated AI $\times$ non-sarcastic texts. The discordant-pair counts are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for the interaction-projection dual-head model vs. the RCNN dual-head model (all $p < .001$). The discordant counts and absolute rate differences provide the substantive effect-size information.

Table 10 gives the language-stratified subgroup rates and Table 11 the corresponding per-language F1 scores. In both architectures and both language strata, the false-positive rate is higher under the joint procedure than under its matched sarcasm-only control. Language remains coupled with corpus, genre, and annotation setting, so the stratified results do not isolate a language effect.

Bootstrap 95% intervals for the RCNN dual-head sarcasm F1 scores in Table 11 are [.646, .680] on the full set, [.802, .836] in Chinese, [.399, .458] in English, [.605, .653] on the human subset, [.748, .798] on Human $\times$ zh, and [.355, .440] on Human $\times$ en.

The final RCNN dual-head model’s MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the human-only subset contains no

AI-authored examples, an MGTD F1 score on that subset would not measure source discrimination; Table 8 therefore reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). This contrast also reflects differences in prevalence, genre, and annotation procedure: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values should therefore not be interpreted as an isolated language effect.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629 on human-authored texts alone, because the overall score also includes AI-generated texts. This point estimate differs from the five-run mean in Table 1, which aggregates fold-based runs. Within the human subset, absolute joint-versus-sarcasm-only differences in sarcasm F1 are below 0.03 and vary in direction. The negative-transfer effect is therefore concentrated in the subgroups containing AI-generated texts, consistent with §5.2.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore lies mostly above the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

The coexistence of an upward score shift and substantial within-AI discrimination motivates the representation analyses in Appendix E.

C.4 Evaluation protocol, inter-judge rank correlations, and condition-level tests (§6). Each sample is evaluated separately by each of the three judge APIs in one API call using the same prompt structure for all model conditions (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source label. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. The ratings are therefore obtained without ground-truth source labels, although the judges are explicitly asked to infer source before assigning the two ratings. Base and SFT outputs are

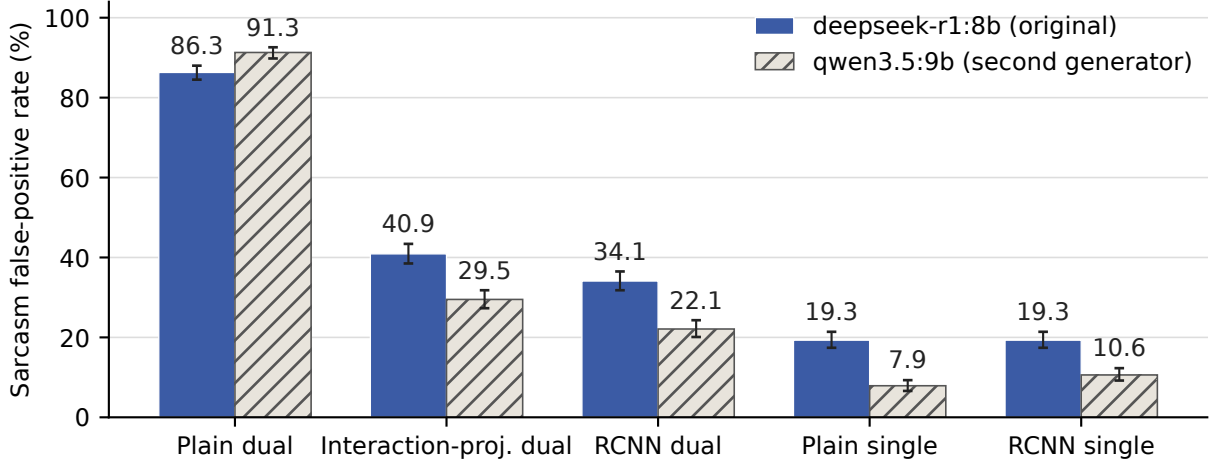


Figure 2: Sarcasm false-positive rates on AI-generated texts prompted to be non-sarcastic, for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

Model	Lang.	<i>n</i>	Dual FP; rate [CI]	Single FP; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.1]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 10: Language-stratified sarcasm false-positive rates on AI-generated texts prompted to be non-sarcastic (%; Wilson 95% intervals). Dual FP and Single FP give the number of false positives under the dual- and single-head models; Δ is the dual rate minus the single rate. Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

evaluated separately. The judges were accessed through the model identifiers `deepseek-chat`, `gemini-2.5-flash-lite`, and `gpt-5-nano`. At the time of our May 2026 API calls, DeepSeek’s April 24, 2026 API announcement identified `deepseek-chat` as routing to the non-thinking mode of DeepSeek-V4-Flash; the legacy identifier was subsequently retired on July 24, 2026 (DeepSeek, 2026).

Pairwise inter-judge rank correlations are low to moderate, with Spearman ρ ranging from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). Table 13 reports the corresponding condition-level tests. We therefore do not treat individual-text rankings as a common gold standard. The condition-level observation rests separately on the consistent rating direction across all six setting–judge comparisons.

Outputs from the SFT model receive lower mean sarcasm ratings in all six setting–judge comparisons, and Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that five differences remain significant. The pooled correlations in Ta-

ble 12 use the three-judge mean as a descriptive aggregate, while the limited pairwise rank correlations make judge-specific results necessary. Four instances each have one unparseable rating but retain two valid judge ratings, so averaging the available ratings preserves all 600 outputs in the aggregate.

C.5 Paired tests between SFT outputs and human-authored reference texts (supplementary). These tests are performed on the Chinese subset of the 300-prompt generation evaluation ($n = 180$; three GPT-5-nano ratings are missing, leaving 177). The comparisons provide judge- and dimension-specific comparison context: for DeepSeek-V4-Flash, SFT outputs receive significantly lower human-likeness ratings than the reference texts (-0.161 , $p = 0.0025$), whereas for GPT-5-nano, SFT outputs receive significantly higher sarcasm ratings than the reference texts ($+0.186$, $p = 0.0114$). The remaining four comparisons do not reach significance at the reported threshold. Thus, the observed reference differences vary across judges and dimensions; these tests do not estimate a judge-by-dimension interaction.

Model	Full set	Chinese	English	Human subset	Human $\times$ zh	Human $\times$ en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
IP dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663	0.819	0.429	0.629	0.774	0.399
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 11: Per-language and human-subset sarcasm F1 (final models). IP denotes the interaction-projection configuration. Bootstrap 95% intervals for the RCNN dual-head row are given in the text below.

Subset	n	Sarc. ρ	Human-like ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 12: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT): Sarc. ρ compares $P(\text{sarcastic})$ with sarcasm ratings, and Human-like ρ compares $P(\text{human})$ with human-likeness ratings. * significant; the other entries are not significant.

Judge	Lang.	n	Base	SFT	adj. p
DeepSeek-V4-F	en	120	4.025	3.292	<.0001
DeepSeek-V4-F	zh	180	4.044	3.878	.0008
Gemini-2.5-FL	en	120	3.992	3.750	.0008
Gemini-2.5-FL	zh	180	4.028	3.994	.2008*
GPT-5-nano	en	119	4.084	3.681	<.0001
GPT-5-nano	zh	180	3.911	3.517	<.0001

Table 13: Condition-level tests for sarcasm ratings obtained without ground-truth source labels (Wilcoxon + BH-FDR). * not significant. Averaging the three Chinese judge means gives 3.99 for base and 3.80 for SFT, as reported in §6. GPT-5-nano lacks one English base rating (C.4: $n = 119$) and three Chinese reference ratings (affecting C.5, not the C.4 Chinese row); tests use pairwise deletion.

C.6 Plain-detector control. With the plain detector, the Chinese sarcasm probability changes from 0.765 to 0.763 (−0.001), the English sarcasm probability changes from 0.843 to 0.659 (−0.184), and predicted human-authorship probability rises by +0.760 in Chinese and +0.705 in English. The two detectors agree on the rise in predicted human-authorship probability. The Chinese sarcasm-probability change is detector-sensitive, whereas both detectors show a decrease in English.

Overall correlations between detector probabilities and corresponding ratings are +0.142 for sarcasm and −0.214 for human-likeness. The corresponding values are −0.037 (not significant) and −0.209 in Chinese, and +0.342 and −0.283 in English. Each differs from the RCNN-

detector value by at most 0.02. Negative associations between predicted human-authorship probabilities and human-likeness ratings are similar across the two detectors, whereas the condition-level sarcasm-probability change differs between the plain and RCNN detectors, especially in Chinese.

C.7 Within-condition correlations (RCNN detector). For predicted human-authorship probability and human-likeness ratings, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese +0.10, English −0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese −0.19, $p = 0.009$; English −0.12, not significant). The pooled value of −0.212 is driven mainly by the opposing condition-level changes: SFT raises predicted human-authorship probability by about 0.67 while lowering the mean rating. The pooled negative correlation should therefore not be interpreted as a strong instance-level inverse relationship. Within either condition, predicted human-authorship probability is not significantly positively correlated with rated human-likeness.

For sarcasm, within-condition correlations are +0.05 for Chinese base (not significant), +0.15 for Chinese SFT ($p = 0.048$), +0.41 for English base, and +0.37 for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.8 Detector probabilities before and after SFT across the two task–language settings. The RCNN detector assigns the following mean probabilities to base and SFT outputs:

Lang.	Cond.	$P(\text{sarcastic})$	$P(\text{human})$
Chinese	Base $\rightarrow$ SFT	0.780 $\rightarrow$ 0.895	0.289 $\rightarrow$ 0.969
English	Base $\rightarrow$ SFT	0.854 $\rightarrow$ 0.730	0.198 $\rightarrow$ 0.864

Table 14: Mean detector probabilities before and after SFT (RCNN detector).

Table 14 reports these means. Sarcasm probabilities move in opposite directions in the Chinese comment-generation and English rewriting settings (+0.115 and -0.124, respectively), while predicted human-authorship probabilities rise sharply in both. GenTrain itself is balanced at 1,214 instances per language, but the evaluation settings differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast.

The mean LLM-judge ratings for the human-authored reference texts are 3.77 for sarcasm and 4.17 for human-likeness; the corresponding values are 4.01 and 4.24 for the base model and 3.71 and 4.08 for SFT. From the base model to SFT, the mean human-likeness rating decreases even as predicted human-authorship probability increases, as analyzed in §6.

D Illustrative Cases of Disagreement between Detector Probabilities and LLM-Judge Ratings

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector probabilities and ratings from the three LLM judges to show large disagreements; no formal ranking or top- k selection was performed. The selection is non-systematic, and the cases should not be used to estimate the prevalence of any disagreement type. Detector probabilities range from 0 to 1, and LLM-judge ratings are averaged across the three evaluators on a 1–5 scale. Long texts are shortened for display, while Chinese examples retain the original model input and provide Hanyu Pinyin transliterations and English translations in brackets.

The three disagreement types (Table 15, Table 16, and Table 17) show, respectively, high predicted human-authorship probabilities with low LLM-judge ratings of human-likeness, high predicted sarcasm probabilities with low LLM-judge ratings of sarcasm, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with an obvious reasoning-style preface is nevertheless assigned predicted human-authorship probability 1.00 by the MGTD head, corresponding to the disagreement discussed in §6.

Together, these cases motivate candidate hypotheses involving surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences, as well as semantic reversal, whose prevalence and causal effects require

Lang.	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [<i>Hao, wo xuyao zhendui Sanxing huidui Huawei de xinwen shengcheng yi tiao fengci pinglun. Shouxian...</i> ; “Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (reasoning-style preface)
Chinese	SFT	1.00	2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [<i>Paofen jingsai: canshu jiaolv: paofen jiaolv: shuihuo: hahaha</i> ; “Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
English	SFT	1.00	2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 15: Cases with high predicted human-authorship probability but low LLM-judge ratings of human-likeness. The detector column reports $P(\text{human})$.

systematic testing. The examples illustrate several disagreement types compatible with the near-zero aggregate Chinese sarcasm correlation in §6.

E Representation-Level Analysis of Error Distribution between Source Groups

This appendix specifies the representation sources, statistical inputs, diagnostic operators, and readouts used in §5.4.

E.1 Analysis overview and representation sources. The primary analysis uses fixed pooled vectors from the frozen final plain joint model behind the 86.3% result. Its matched sarcasm-only model is used only for the baseline source-decodability comparison. Full-set source probes and logit correlations use all applicable DualTest examples. Interventions use a fixed StratifiedShuffleSplit (test size 0.5, random state 42) stratified by the four source ×

sarcasm subgroups. Fitting/evaluation counts are 411/412 for human $\times$ sarcastic, 774/775 for human $\times$ non-sarcastic, 390/389 for AI $\times$ sarcastic, and 757/756 for AI $\times$ non-sarcastic. The halves contain 964/934 Chinese and 1,368/1,398 English texts. Human-AI pair links are not used, and 24.7% cross halves.

Label-dependent probe directions, INLP subspaces, LEACE, and the INLP stage used to construct the study-specific INLP-3 residual PC1 are estimated on the fitting half. Some operators also use unlabeled full-set statistics: covariance-aware alignment uses the covariance of all DualTest representations, and the task-head, source-probe, INLP, and Euclidean random removals use their pooled mean. LEACE, covariance-whitened controls, and INLP-3 residual PC1 instead use fitting-half statistics. Fixed transformations are applied to evaluation representations, which are read by the unchanged sarcasm head and by newly

Lang.	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sanxing: wo cai shi diyi! Ni kankan wo zhe shexiangtou zongfen, 109+96, 112+89, wo ying le!; “Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
Chinese	SFT	0.99	2.67	“恭喜恭喜，美国牛牛牛，继续加油” [Gongxi gongxi, Meiguo niu niu niu, jixu jiaoyou; “Congratulations, congratulations, America is awesome awesome, keep it up”] (formulaic mock-praise)
English	SFT	0.93	2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 16: Cases with high predicted sarcasm probability but low LLM-judge ratings of sarcasm. The detector column reports $P(\text{sarcastic})$.

cross-validated source probes. Figure 3 summarizes these branches and readouts.

E.2 Probe, direction, and statistical protocols.

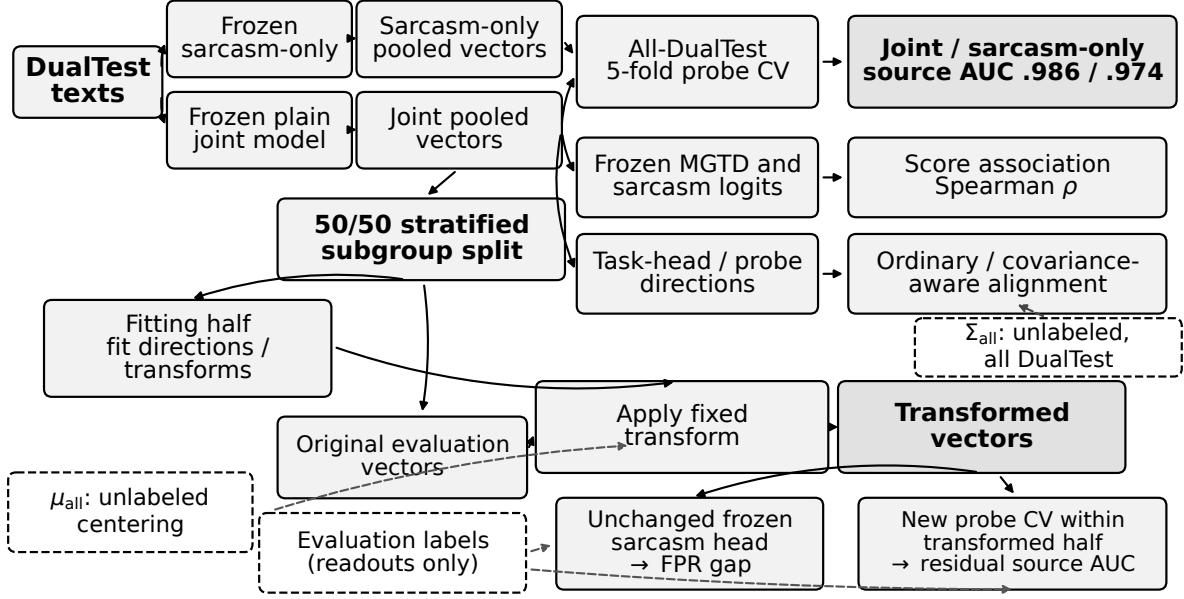
Table 18 separates frozen representation sources, label-dependent estimation, unlabeled statistics, and evaluation readouts. Source probes use scikit-learn logistic regression with the lbfgs solver, ℓ_2 penalty, $C = 1.0$, no class weighting or feature scaling, and tolerance 10^{-4} . AUC uses cross_val_score with five-fold stratified cross-validation, no shuffling, and ROC AUC scoring. Ordinary probes use at most 2,000 iterations; each INLP step uses at most 1,500.

Each task head is Dropout followed by

Lang.	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	Base	0.14	4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [Telangpu chengnuo 120 yi butie xiang sa dabing, nongmin shoudao de que shi suoshui cheng yingbi de chengnuo...; “Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
Chinese	Base	0.24	4.00	“特斯拉：价格跳水后又跳水。消费者：钱包跳水后又跳水。” [Tesila: jiage tiaoshui hou you tiaoshui, xiaofeizhe: qianbao tiaoshui hou you tiaoshui.]; “Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
English	Base	0.12	4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 17: Cases with low predicted sarcasm probability but high LLM-judge ratings of sarcasm. The detector column reports $P(\text{sarcastic})$.

(a) Representation-diagnosis protocol



(b) INLP-3 residual PC1 construction and evaluation

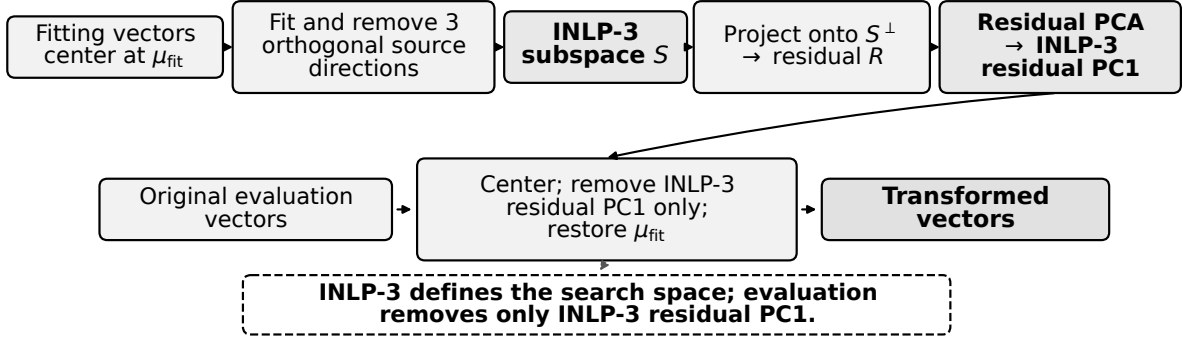


Figure 3: Representation-diagnosis protocol. (a) Full-set probes are separate from split-sample interventions. Label-dependent directions and transformations use the fitting half, while covariance-aware alignment uses the full-set unlabeled covariance and specified Euclidean removals use the full-set unlabeled mean. Transformed evaluation vectors are read by the unchanged sarcasm head for source-group false-positive-rate gaps and by newly cross-validated source probes for residual linear source AUC. (b) INLP-3 defines the search space for the study-specific INLP-3 residual PC1; evaluation removes that PC1 alone from the original representations, not the INLP-3 subspace.

Linear(1024,2). For a binary linear head, the class-weight-row difference d is its logit-contrast direction: for representation z , the positive-minus-negative logit has the form $d^T z + \Delta b$, so moving z along d increases that contrast. The task-head directions are $d_{\text{sarc}} = W_s[1] - W_s[0]$ for the sarcastic-minus-non-sarcastic logit and $d_{\text{AI}} = W_a[0] - W_a[1]$ for the AI-minus-human logit (MGTD class 1 denotes human). Directional analyses omit bias; predictions use the complete head with dropout disabled in evaluation mode. For

comparison, the fitting-half source probe supplies an independently estimated source direction: its ℓ_2 -normalized logistic-regression coefficient, oriented toward human authorship.

Ordinary cosine compares directions in Euclidean weight space. Covariance-aware alignment is $\cos(\Sigma_{\text{all}}^{1/2} w_1, \Sigma_{\text{all}}^{1/2} w_2)$, where Σ_{all} is the Ledoit–Wolf covariance (Ledoit and Wolf, 2004) of all pooled DualTest representations with eigenvalues clipped below 10^{-8} . This use of all unlabeled representations makes the alignment

Component	Frozen source	Object derivation	Unlabeled statistics	Application / evaluation	Readout or role
Full-set joint probe	Final plain joint vectors	Five-fold source-probe CV on all DualTest	None additionally specified	Held-out fold in full-set CV	Baseline source AUC 0.986
Full-set sarcasm-only probe	Matched sarcasm-only vectors	Same full-set CV protocol	None additionally specified	Held-out fold in full-set CV	Comparison AUC 0.974
Logit association	Final plain joint model	None	None	AI $\times$ non-sarcastic texts	Spearman correlation
Task-head directions	Frozen joint task heads	Read $W_s[1] - W_s[0]$ and $W_a[0] - W_a[1]$	Σ_{all} for alignment; μ_{all} for Euclidean removal	Alignment; removal from evaluation vectors	Alignment; post-removal FPR gap and residual AUC
Fitting-half probe direction	Joint fitting vectors	Logistic coefficient from fitting-half source labels	Σ_{all} for alignment; μ_{all} for removal	Alignment; removal from evaluation vectors	Probe-head alignment; post-removal FPR gap and residual AUC
INLP	Joint fitting vectors	Iterative source probes with orthogonalization	μ_{all}	Remove $k \in \{1, 2, 4, 8, 16\}$ directions from evaluation vectors	Multi-direction Euclidean diagnostic; FPR gap and residual AUC
LEACE	Joint fitting vectors	Fitting vectors and source labels	$\mu_{\text{fit}}, \Sigma_{\text{fit}}$	Fixed transform on evaluation vectors	Covariance-aware diagnostic; FPR gap and residual AUC
INLP-3 residual PC1	Joint fitting vectors	Labels fit INLP-3; residual PCA is unlabeled	μ_{fit} and residual covariance	Remove PC1 only from original evaluation vectors	Residual-geometry diagnostic; FPR gap and residual AUC
Post-intervention probe	Transformed evaluation vectors	Five-fold source-probe CV within each transformed set	None additionally specified	Held-out folds within transformed evaluation half	Residual source AUC

Table 18: Probe, direction-estimation, statistical, and post-intervention evaluation protocols. μ_{all} and Σ_{all} use all unlabeled DualTest representations; μ_{fit} and Σ_{fit} use the fitting half. Directional analyses omit the task-head bias, whereas predictions use the complete frozen head. Euclidean random controls use μ_{all} ; covariance-whitened controls use fitting-half statistics. Object derivation states how each analysed object is obtained and whether source labels are used. Permuted-label and LEACE in-sample checks appear in E.5.

transductive. Signed values are -0.41197 for the source-probe versus sarcasm directions and $+0.48713$ for the MGTD versus sarcasm directions; the main text reports absolute values because source-direction orientation changes the sign.

E.3 Intervention operators. For representations X , selected mean μ , and orthonormal direction matrix U , centered Euclidean removal is

$$\mathcal{E}_\mu(X, U) = X - [(X - \mu)U]U^\top, \quad (3)$$

which subtracts the selected component while retaining the mean. Task-head, source-probe, INLP, and Euclidean random removals use μ_{all} ; LEACE, covariance-whitened controls, and INLP-3 residual PC1 use μ_{fit} .

INLP (Ravfogel et al., 2020) repeatedly fits a

source probe to the current fitting residuals, normalizes and Gram–Schmidt-orthogonalizes its coefficient, and deflates the representation. We evaluate $k \in \{1, 2, 4, 8, 16\}$ accumulated directions. Same-dimensional Euclidean random subspaces use QR-orthogonalized Gaussian matrices. These controls and permuted labels use `default_rng(0)`.

LEACE (Belrose et al., 2023) is implemented directly. On the fitting half, let Σ_{fit} be the Ledoit–Wolf covariance with eigenvalues clipped below 10^{-8} , and let $S_{xz} = X_c^\top Z_c/n$ for centered representations X_c and centered one-hot source labels Z_c . We compute $M = \Sigma_{\text{fit}}^{-1/2} S_{xz}$ and retain singular directions with $s > 10^{-6}$, giving rank one. With whitened direction matrix Q , $P = \Sigma_{\text{fit}}^{1/2} Q Q^\top \Sigma_{\text{fit}}^{-1/2}$ and erasure maps X to $X - (X -$

$\mu_{\text{fit}})P^\top$. Ten covariance-whitened controls sample unit Gaussian q and use $\Sigma_{\text{fit}}^{1/2}qq^\top\Sigma_{\text{fit}}^{-1/2}$.

For the exploratory, study-specific INLP-3 residual PC1, three fitting-half INLP directions define a low-dimensional source subspace S . The residual matrix is

$$R = (X_{\text{fit}} - \mu_{\text{fit}}) - [(X_{\text{fit}} - \mu_{\text{fit}})S]S^\top. \quad (4)$$

The largest-eigenvalue vector of $\text{np.cov}(R^\top)$ defines INLP-3 residual PC1. Evaluation applies Equation 3 to remove this single axis from the original evaluation vectors; the three INLP directions define the search space but are not themselves removed at evaluation time.

E.4 Diagnostic and intervention evidence.

This appendix records the full chain summarized in §5.4. Q1 asks whether source is available in the representation, Q2 whether source-related quantities are associated with the sarcasm decision, and Q3 whether interventions change how the frozen sarcasm head allocates false positives across source groups. Source is highly linearly decodable: full-set AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts; the matched sarcasm-only model yields 0.974. On AI $\times$ non-sarcastic texts, MGTD AI and sarcasm logits correlate at $\rho = 0.519$, with length-tertile correlations of 0.44, 0.56, and 0.63. Task-head directions are nearly orthogonal under ordinary cosine ($\cos \approx -0.03$), whereas covariance-aware absolute alignment is 0.49 for the MGTD-head direction and 0.41 for the fitting-half source-probe direction. These results associate source-related scores and data-weighted directions with the affected decisions.

For interventions, gap is AI minus human FPR on non-sarcastic evaluation texts; source AUC is obtained from a new probe cross-validated within each transformed evaluation set. The untouched baseline is 84.788% AI FPR and 32.774% human FPR (gap 52.014 pp), with source AUC 0.98197.

Table 19 summarizes the interventions. The MGTD-head, source-probe, Euclidean-random, and INLP interventions do not narrow the gap; INLP at $k = 16$ widens it to 72.1 pp. Their residual source AUC also remains above 0.97, so these operators do not remove source in the first place, and their effect on the gap is uninformative about whether source matters. LEACE lowers measured source AUC to 0.589 and the gap to 19.1 pp; no

whitened random direction produces a comparable change. These controls nevertheless perturb representations far less than LEACE does—mean relative change 0.02 against 0.53—so they establish only that an arbitrary whitened direction is insufficient and do not separate erasing source from perturbing the representation strongly. The INLP-3 residual PC1 condition supplies that separation: its mean relative change is 0.57, comparable to LEACE, yet it leaves source AUC at 0.982. The unweighted mean FPR of the two source groups is nearly unchanged ($58.8 \rightarrow 59.1$) because the human non-sarcastic rate rises from 32.8 to 49.5, showing error reallocation rather than lower average error.

Removing INLP-3 residual PC1 changes AI FPR to 66.7 and the gap to 12.6 pp while source AUC remains 0.982. The new probe can recover source from the remaining representation even though removing this axis reallocates errors across the two source groups. Together with the LEACE result, this shows that the FPR gap is not determined by residual linear source AUC alone: both covariance-aware source structure and a residual high-variance axis can affect how the fixed sarcasm head distributes errors between source groups.

E.5 Implementation checks. Manually computed logits match model outputs with maximum error 1.4×10^{-6} , and projection residuals are about 10^{-6} . LEACE leaves a fitting-half class-mean residual of 7×10^{-6} and fitting-half probe AUC 0.514; these are implementation checks, not held-out results. A permuted-label full-set probe gives AUC 0.512. Thus, the reported AUCs refer to distinct settings: 0.986 is full-set five-fold baseline decodability, 0.982 is the untouched intervention-half baseline, 0.514 is the LEACE in-sample check, and 0.589 is held-out post-LEACE decodability.

Erasure scheme	AI×n FPR	H×n FPR	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
MGTD-head direction	86.2	30.2	56.0	0.982
Source-probe direction	84.8	32.9	51.9	0.981
Euclidean random direction	85.2	32.6	52.5	0.982
INLP ($k = 16$)	89.8	17.7	72.1	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Whitened random × 10 (mean)	84.9	32.7	52.2	0.982
INLP-3 residual PC1	66.7	54.1	12.6	0.982

Table 19: Intervention summary. H×n = human × non-sarcastic; AI×n = AI × non-sarcastic; FPR in %; gaps use unrounded rates. Whitened-control values average ten directions, whose gaps range from 51.4 to 52.7 pp. AUC is the newly cross-validated post-intervention source-probe AUC.